Calculus I

Tutorial #10 (02–06 Dec, 2024)

Topics:

- Trigonometric substitution
- Integrals of rational functions
- Improper integrals

Problem 1. Evaluate the integrals

a)
$$I = \int_0^{3/2} \frac{dx}{\sqrt{9 - x^2}}.$$

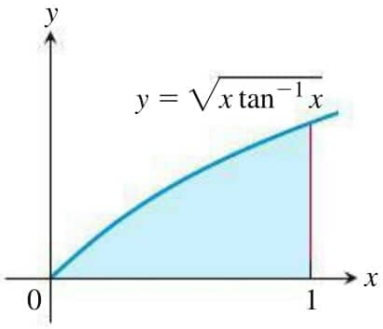
a)
$$I = \int \sqrt{1 - 9t^2}\, dt.$$

Problem 2. Use any method to evaluate the integrals. Most will require trigonometric substitutions, but some can be evaluated by other methods.

a)
$$I = \int \sqrt{\frac{x+1}{1-x}}\, dx.$$

b)
$$I = \int \frac{v^2\, dv}{(1 - v^2)^{5/2}}.$$

Problem 3. Consider the region bounded by the graphs of $y = \sqrt{x \arctan x}$ and $y = 0$ for $0 \leq x \leq 1$. Find the volume of the solid formed by revolving this region about the x-axis (see the accompanying figure).



Problem 4. Evaluate $I = \int x^3\sqrt{1-x^2}\,dx$ using

(a) Integration by parts

(b) An u-substitution

(c) A trigonometric substitution.

Problem 5. Express the integrand as a sum of partial fractions and evaluate the integral

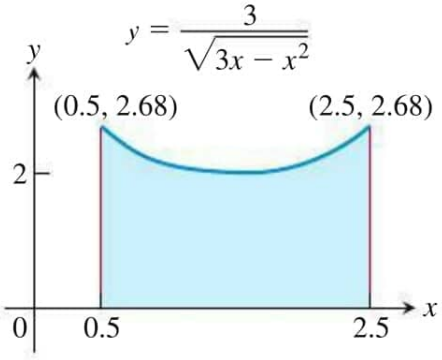
a)
$$\int \frac{dt}{t^3+t^2-2t}.$$

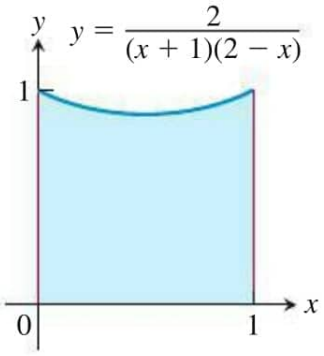
b)
$$\int \frac{2s+2}{(s^2+1)(s-1)^3}\,ds.$$

Problem 6. Find the volume of the solid generated by revolving the shaded region about the indicated axis.

a) The x–axis

b) The y-axis





Problem 7. The integrals converge. Evaluate the integrals without using tables.

a)

$$I = \int_0^\infty 2e^{-\theta} \sin\theta \, d\theta.$$

b)

$$I = \int_0^2 \frac{dx}{\sqrt{|x - 1|}}.$$

Problem 8. Use integration, the Direct Comparison Test, or the Limit Comparison Test to test the integrals for convergence. If more than one method applies, use whatever method you prefer.

a)

$$\int_0^\infty \frac{d\theta}{1 + e^\theta}.$$

b)

$$\int_{e^e}^\infty \ln(\ln x) \, dx.$$

Problem 9. Evaluate the integral

$$\int \frac{dx}{x(x+1)(x+2)\cdots(x+m)}.$$

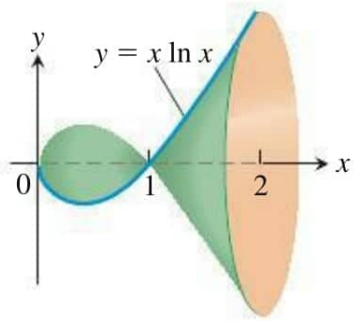
Problem 10. (*Finding volume*). The region between the x-axis and the curve

$$y = f(x) = \begin{cases} 0, & x = 0 \\ x \ln x, & 0 < x \leq 2 \end{cases}$$

is revolved about the x-axis to generate the solid shown here.

(a) Show that f is continuous at $x = 0$.

(b) Find the volume of the solid.



THE END

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Answer to selected problems

3. (a) $\frac{\pi}{6}$ (b) $\frac{\arcsin(3t) + 3t\sqrt{1-9t^2}}{6} + C$
5. (a) $\frac{\ln|t-1|}{3} - \frac{\ln|t|}{2} + \frac{\ln|t+2|}{6} + C$ (b) $\arctan(s) + \frac{1}{(s-1)} - \frac{1}{(s-1)^2} + C$
6. (a) ≈ 30.34 (b) ≈ 2.9
7. (b) 4
8. (a) converges
10. (b) ≈ 2.015